

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CE) Examination

Nov-Dec 2013

CE202: DATA COMMUNICATION & NETWORKING

Date: 23/11/2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Do as Directed.

[07]

- (1) Define Redundancy with respect to Error Control.
- (2) Cable TV technology is example of _____ topology.
- (3) In which topology, n devices in a network requires n cable links?
- (4) Radio Communication frequencies range from _____ KHz to _____ GHz.
- (5) What do mean by Spike in Impulse Noise?
- (6) Calculate Propagation Time for Twisted Pair Cable.
- (7) Imagine a signal travels through a transmission medium and its power is reduced to half. How much power loss is occurred here?

Q - 2 (a) Calculate redundancy bits on data 1001101 using Hamming Code.

[02]

- (b) Draw and explain topology which uses more than one hub. Also write down pros and cons of that topology. [03]
- (c) Explain factors that affect the performance of the network. [04]
- (d) Write down advantages and disadvantages of Optical Fiber. [05]

OR

Q - 2 (a) How to know whether data having values 10101001 00111001 and checksum having value 00011101 contains error or not? [02]

- (b) Which are the key elements of Protocol? Explain them. [03]
- (c) Explain different reliability and security parameters for effective and efficient network. [04]
- (d) Compare performance of different transmission media. [05]

Q - 3 (a) Compare & contrast Hub and Switch. [02]

- (b) Explain 802.1 and 802.2 (both layers) of Project 802. [03]
- (c) What is a need for Encryption and Decryption? Explain with appropriate diagram. [04]
- (d) Compare Ethernet, Fast Ethernet, Gigabit Ethernet, Token Ring and FDDI. [05]

OR

Q - 3 (a) Explain any two switching methods. [02]

- (b) Explain Switched Ethernet and Gigabit Ethernet. [03]
- (c) What is MHS in application layer? Explain message format in MHS. Also write down need for MHS. [04]

(d) Draw and explain 802.3 MAC frame.

[05]

SECTION - II

Q - 4 (a) Do as Directed.

[07]

- (1) A digital signal has bit interval of 40 microseconds. Calculate bit rate.
- (2) Define Harmonics.
- (3) We want to digitize human voice. What is the bit rate assuming eight bits per sample?
- (4) Consider audio signal with a bandwidth of 4 MHz. Calculate bandwidth required to modulate same signal with FM.
- (5) Define interface.
- (6) A sine wave is offset $1/6$ of a cycle with respect to time zero. The resulting phase is _____ degrees.
- (7) AM stations are allowed carrier frequencies anywhere between _____ KHz to _____ KHz

Q - 5 (a) Which characteristics are used to describe Sine Wave? Explain any two of them.

[02]

(b) Define these three terms: Encoding Digital data into Digital signal, Digitizing an analog signal, modulating a Digital signal.

[03]

(c) Explain Cable Modem in detail.

[04]

(d) What do you mean by DTE-DCE interface? Explain it in detail.

[05]

OR

Q - 5 (a) Explain Periodic and Aperiodic signals.

[02]

(b) Define these three terms: NRZ, RZ and Biphas.

[03]

(c) Explain 56K Modem in detail.

[04]

(d) Explain Synchronous Transmission and Asynchronous Transmission in Detail.

[05]

Q - 6 (a) Compare Go-Back-n and Selective-Reject.

[02]

(b) Explain any three weaknesses of Circuit-Switching.

[03]

(c) Explain Stop-and-Wait along with pros and cons.

[04]

(d) Explain three keys to understand Distance Vector Routing. Draw and explain format for routing table in this algorithm.

[05]

OR

Q - 6 (a) What do you mean by Line Discipline? Explain in short.

[02]

(b) Explain example of circuit-switched telephone network.

[03]

(c) Explain Sliding Window in detail.

[04]

(d) Explain three keys to understand Link State Routing. Draw and explain format for link state packet in this algorithm.

[05]